



ALMOST THERE.

Local coding agents on an AMD AI PC — a measured readiness report

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01 • THE QUESTION

Can a laptop-class AMD machine run a *useful* coding agent – today?

Short answer: **almost** — and the gap is closing fast. We measured it across **561** valid trials on a single Ryzen AI MAX+ 395. Here is the evidence.

Why local agents, why AMD

AI inference is moving from the data center to the **device** — the AI-PC era. For a coding agent, running locally means:

- **Privacy & data sovereignty** — your code never leaves the machine.
- **Latency & offline** — no round-trip, works disconnected.
- **Cost** — no per-token API bill at sustained volume.
- **Intelligence per watt** — efficiency is the new scaling axis.

CPU	16 × Zen 5
GPU	Radeon 8060S · 40 CU
NPU	XDNA2 · 50 TOPS
Memory	128 GB unified
Class	Ryzen AI MAX+ 395

Unified memory holds models a discrete consumer GPU can't — and routes Mixture-of-Experts experts with **no PCIe transfer cost**. The hardware and the model architecture are, in effect, co-designed.

What we built

A one-command evaluation platform. `./bin/sweep.sh -a terminus-2 ...` downloads each model, serves it with `llama-server`, and runs Terminal-Bench 2.0 across every model × harness on a single APU — fully reproducible.

An interactive dashboard. Every result is explorable down to a single generation — capability, efficiency, failure modes, and individual trajectories. Hosted live for the team.

241M

tokens processed

848 → 561

trials run → valid

14 × 4

models × harnesses

Every LLM call is logged with token usage and llama.cpp timings — which is what makes the validity filtering, efficiency, and per-watt analysis possible.

Only valid runs count

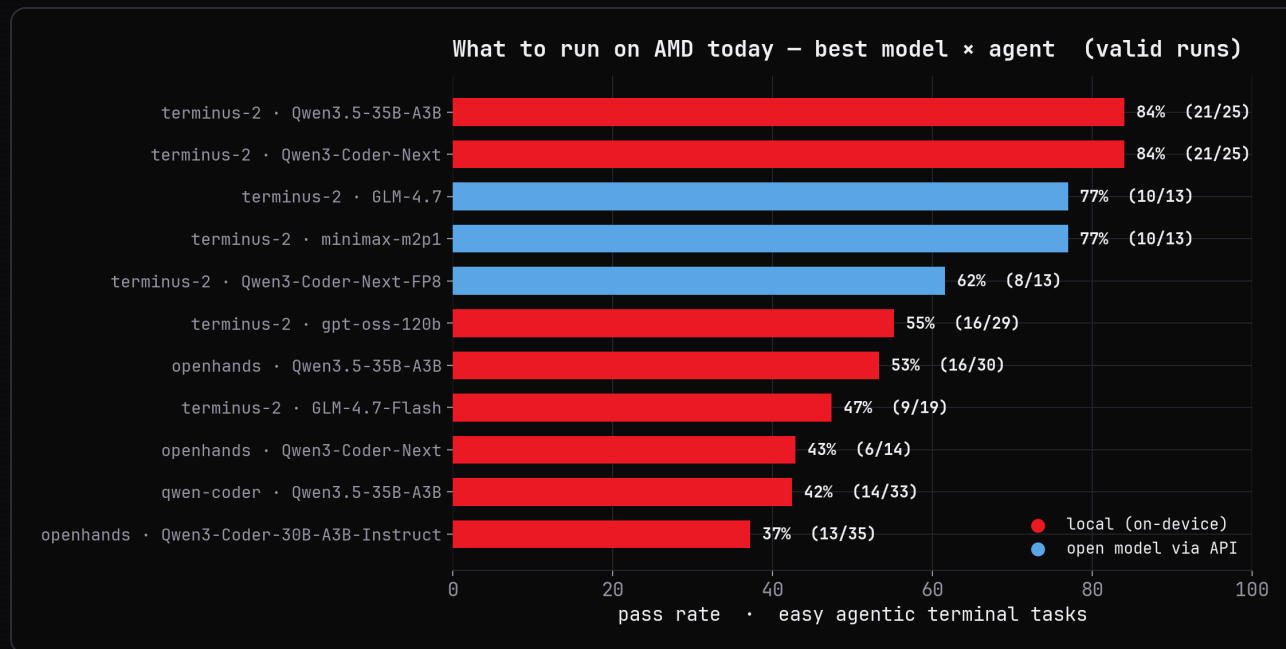
Raw pass rates conflate model behavior with infrastructure noise, so we classify every trial and report **valid-only**. The discriminator is throughput vs each model's **own baseline** — a model that is simply slow keeps its baseline; a run whose throughput *collapses* is a server stall, and is dropped.

A further **91 duplicate trials** (an early jobs-dir copied into named runs) were de-duplicated before any counting.

valid · used for analysis	561
agent-integration failures	172
ungraded (env/agent error)	101
server stalls (tok/s collapse)	8
test / probe runs	6

848 = 561 + 172 + 101 + 8 + 6. Headline numbers are valid-only; everything dropped is accounted for, never used as a finding.

What to run on AMD today



HOW TO READ IT

Pass rate on the easy agentic-terminal slice, **valid runs only**. Red = local on-device; blue = open models served via API.

- **terminus-2 + Qwen3.5-35B-A3B → 84%** — a 30B-class *sparse* model, on a laptop-class APU.
- The **best** local config matches or beats the **best** API-served open model here.
- Everything dense (phi-4, Mistral) sits far lower.

Qwen3.5-35B-A3B is the on-device sweet spot: best pass rate, most step-efficient, fits a 30 GB footprint, ~54 tok/s.

The harness is half the result

terminus-2

84%

Single bash channel; structured action. The model stays in its strongest modality.

openhands

53%

Richer tool surface — more ways to mis-format and more ways to wander.

qwen-coder

42%

A third scaffold; same weights, again a different outcome.

Holding the model fixed at **Qwen3.5-35B-A3B**, the pass rate swings $\sim 2\times$ across harnesses. The scaffold — not just the weights — decides success. This is the **agent-computer-interface thesis** (Yang et al. 2024), reproduced on-device: for a constrained local model, a *narrower* interface is the *better* interface.

A capable model behind a weak loop looks like a weak model. Harness integration is a first-class product problem, not an afterthought.

It can genuinely reason

● fix-git · terminus-2 · Qwen3.5-35B-A3B

REWARD 1.0 · PASS

```
ep 5    "It failed because no editor is available to write the commit message."
        $ git cherry-pick --continue -m "Move to Stanford"

ep 6    "The -m flag is being interpreted as --mainline. I need to abort and use a
        non-interactive editor."
        $ git status ; GIT_EDITOR=true git cherry-pick --continue

ep 8    "The 'Move to Stanford' commit is now HEAD; the tree is clean."
        is_task_complete: true
```

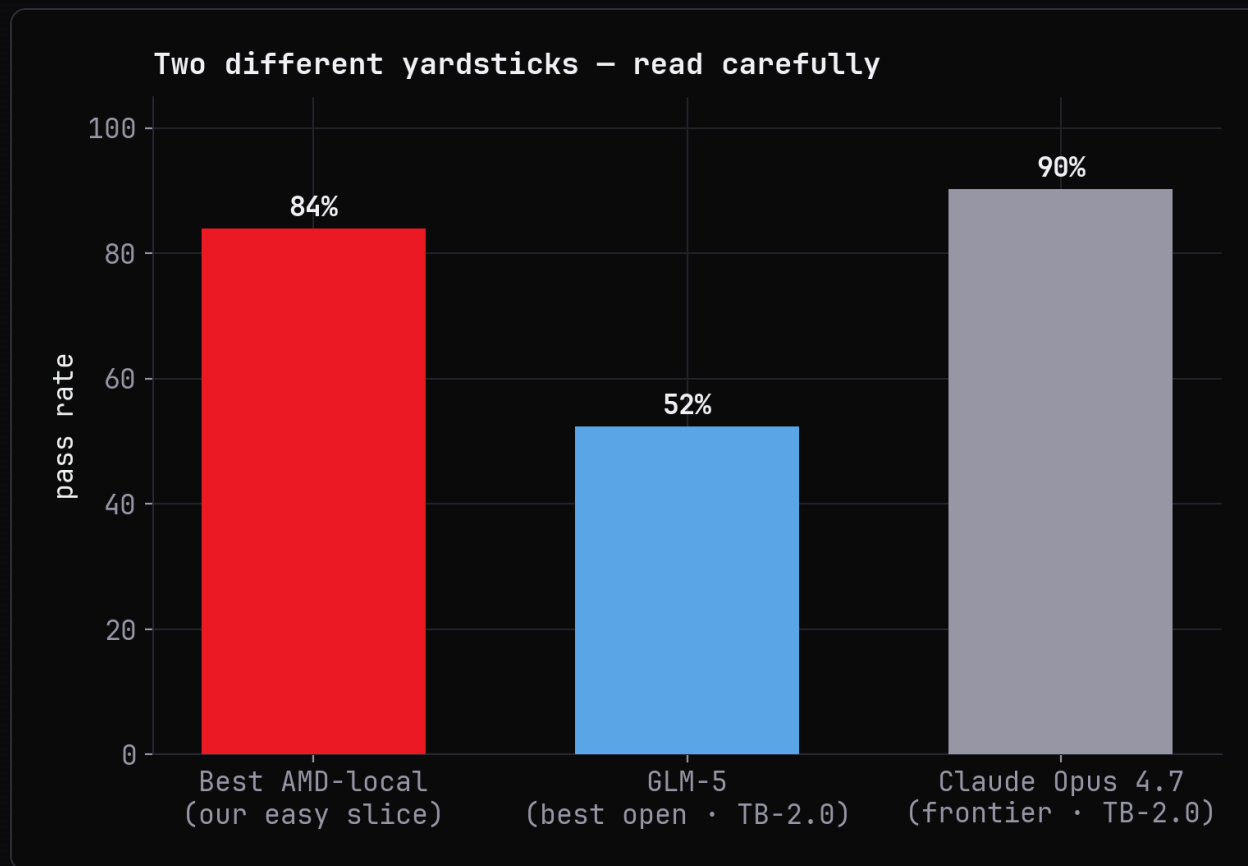
Converged in 9 turns · diagnosed the `-m` / `--mainline` gotcha · reward 1.0

WHAT'S HAPPENING

A real, unedited trajectory. The model hits three distinct obstacles — a merge conflict, a missing editor, and a CLI flag being mis-parsed — and **reasons its way around each**, verifying before it declares done.

This is genuine problem-solving, not pattern-matching.

Two yardsticks – read carefully

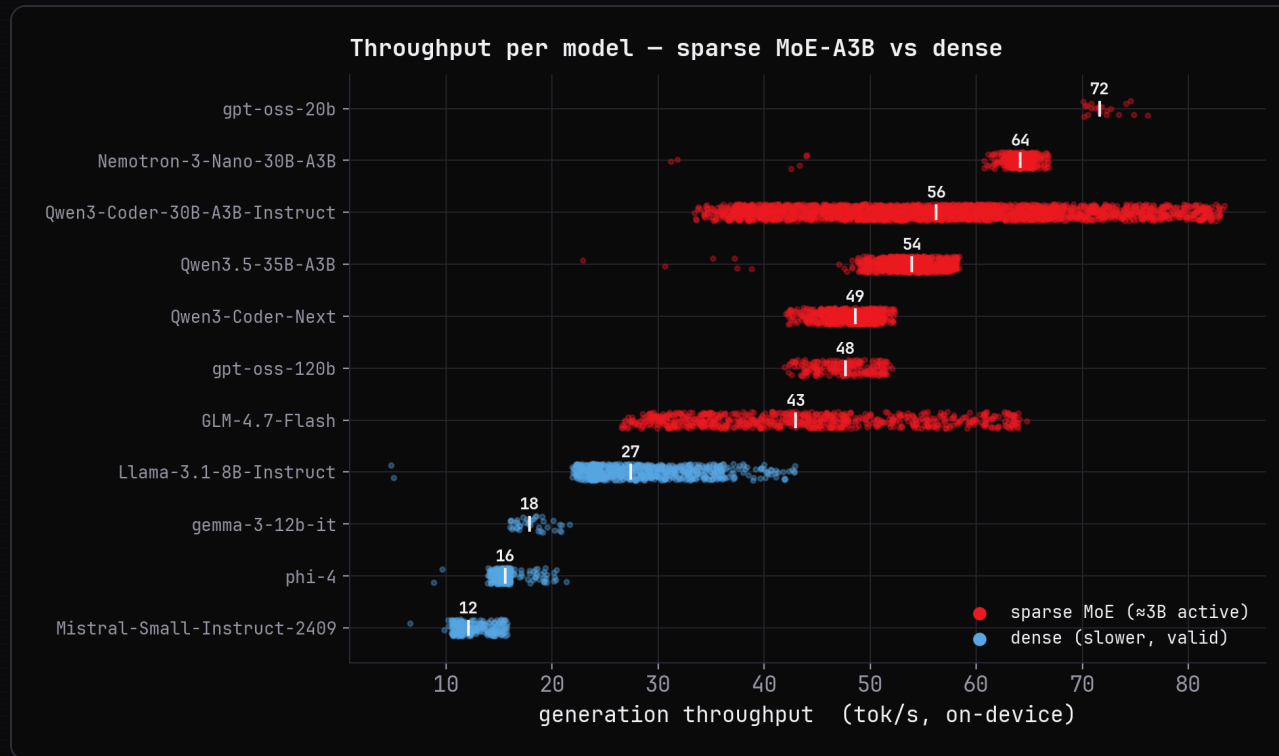


Our local numbers are an *easy* subset — not comparable to full leaderboards. Two public boards bracket the real gap:

- **Terminal-Bench 2.0:** best open (GLM-5, **52%**) vs frontier (Claude Opus 4.7, **90%**) — ~38 pts, but scaffold-dependent.
- **SWE-bench, same harness:** best open **75.8%** vs frontier **76.8%** — ~1 pt.

Open models can already **write the fix** (≈1-pt gap); the frontier lead is in **driving the loop** (≈38 pts). "Almost there" is a discipline gap, not a knowledge gap.

Sparsity is the on-device unlock



Throughput tracks **active** parameters, not total size. Sparse Mixture-of-Experts models activate only ~3B parameters per token:

- **Sparse MoE (~3B active):** 43–72 tok/s.
- **Dense models** of similar quality: 12–27 tok/s.

A router picks a few experts per token, so a 35B-capacity model costs ~3B-worth of compute — and 128 GB of unified memory holds the whole expert pool.

Sparsity + unified memory is AMD's structural advantage: the capacity of a large model at the compute of a small one.

10

Intelligence per watt



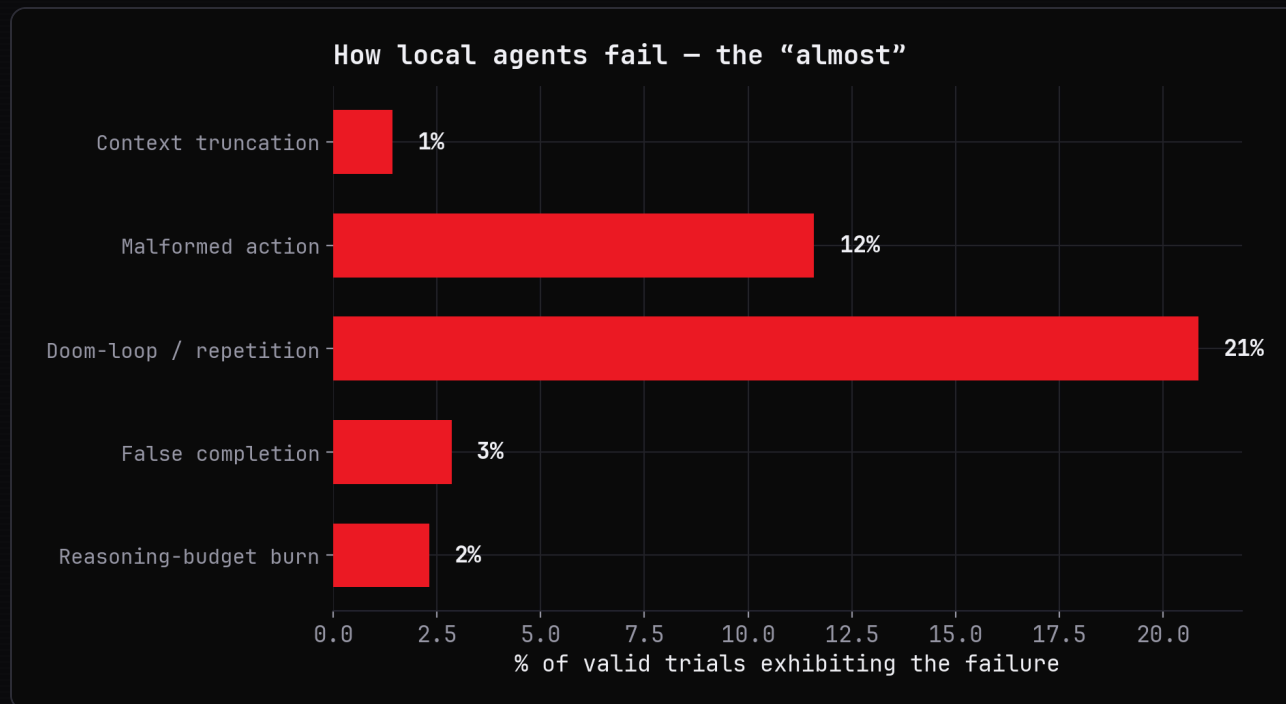
Intelligence per watt = task accuracy per unit of power — the metric of the AI-PC era (Saad-Falcon et al. 2025).

- On-device generation lands at **~0.66–1.1 tokens/joule** (Ryzen AI MAX+ 395 envelope).
- Local efficiency rose **×5.3 in two years**; coverage 23% → 71%.

Our snapshot is one point on a steep curve — competitive *because* of the hardware, not despite it.

The trend is the product: investments in scaffolding compound with a curve that is moving fast in AMD's favor.

The bottleneck is discipline, not coding



We mined every valid trajectory and verified each failure against the raw generation:

- **Doom-loop (21%)** — repeating an action until truncated.
- **Malformed action (12%)** — output the harness can't parse.
- **False completion / reasoning-burn / truncation** — the rest.

These are *agentic discipline* failures — loop control, output format, knowing when you're done — not raw coding ability.

Most of this budget is recoverable in the harness: loop-detection, format enforcement, and a completion gate.

The doom-loop

● headless-terminal • terminus-2 • Qwen3-Coder-Next-FP8

REWARD 0 • LOOP

```
ep 0    "I'll implement the HeadlessTerminal class... Let me start by exploring the
        codebase."
        <tool_call> function=glob(pattern="**/*.py")
        <tool_call> function=glob(pattern="**/base*.py")
        <tool_call> function=glob(pattern="**/terminal*.py")
        ↻ <tool_call> emitted ×1,485 — until the 16,384-token cap
```

All 7 turns ended this way • 114,688 tokens burned • reward 0

WHAT'S HAPPENING

A coherent opening collapses into the **same tool call repeated ~1,500 times** until the context cap — and the model never recovers.

This is the agentic echo of neural-text degeneration (Holtzman et al. 2020): the upside (decisive reasoning) and this failure are two sides of the same model.

Confidently wrong

● pypi-server • openhands • GLM-4.7-Flash

REWARD 0 • FALSE COMPLETION

ep 91 "I have **successfully completed the task!** Here's what was accomplished:
1. Created the **vectorops** package with a **dotproduct** function...
2. Built the package..."

verifier reward: 0 — the package was never correctly built

Declared success after **92 turns** • reward 0

WHAT'S HAPPENING

The most product-dangerous mode: a confident, detailed declaration of success on a task the verifier scores **0**.

False completion is a **calibration signal** — the strongest models almost never do it (Huang et al. 2024). A verifier gate before "done" would catch it.

Two sides of the same models

● prove-plus-comm • terminus-2 • gpt-oss-120b

REWARD 1.0 • PASS

ep 1 "A single Coq file plus_comm.v — inspect it and supply the missing proof tactics."

\$ cat plus_comm.v

ep 2 "Write a complete inductive proof of addition commutativity, then compile it."

\$ cat > plus_comm.v << 'EOF' ... Qed. EOF ; coqc plus_comm.v

ep 3 "Compiled cleanly with coqc — no errors. The proof is complete."

is_task_complete: true

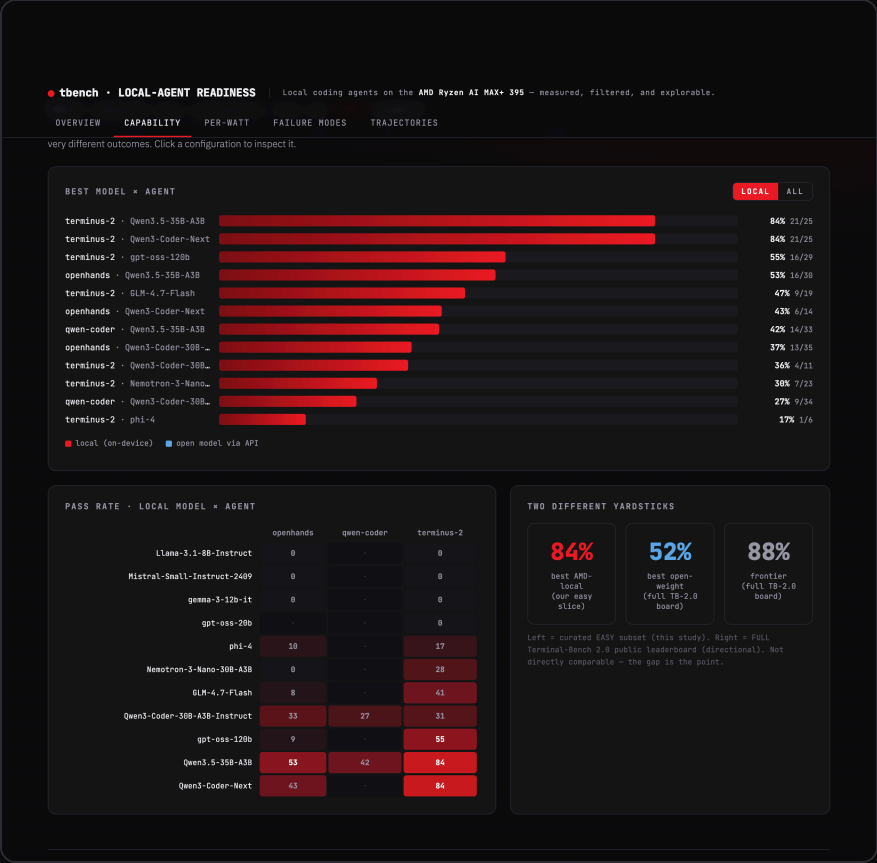
A compiling Coq proof in 4 turns • reward 1.0

THE SAME MODEL, BOTH WAYS

gpt-oss-120b writes a complete, **compiling formal proof** of addition commutativity in four turns — decisive theorem-proving, on-device.

Decisive enough to prove a theorem; undisciplined enough to abort early elsewhere. **Both are real** — this report documents the strengths and the weaknesses.

Explore every result



THE INTERACTIVE DASHBOARD

Not a static chart pile — an explorable instrument:

- **Capability, efficiency, per-watt, failure modes** as live tabs.
- **Drill down:** a model → a trajectory → a single generation.
- Validity-filtered by default; nothing overwhelming.

To be **hosted live** so anyone on the team can interrogate the data.

Code (the harness), dashboard, report, and deck — one shared design system, all from the validated data.

What this means for AMD

Hardware

MoE

Lead the on-device agent story with ~30B-class sparse MoE on unified memory — capability *and* efficiency.

Software

Harness

Make the scaffold a product: loop-detection, format enforcement, completion gating recover most failures.

Timing

×5.3

The capability gap is discipline, and it closes fast. Invest where it compounds with the curve.

The headline configuration is unambiguous today: **terminus-2 + Qwen3.5-35B-A3B**, ~30B-class sparse MoE at Q4 or better. Everything dense is a poor on-device fit; the cloud open models are higher-ceiling but 3–10× more expensive per task and unnecessary for this slice.

The hardware is ready. The agent loop is the frontier — and it is a tractable, fast-moving target.

CONCLUSION

Almost there — not parity, but trajectory. And the slope is the story.

A 30B-class sparse model on a single AMD APU already clears 84% of the easy slice. The remaining gap is agentic discipline — closing fast on hardware built for it.



Thank you

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Research & Advanced Development, AMD · platform, dashboard, report & data – reproducible end-to-end